Auditing Train/Test Leakage in a Polyp Benchmark: Kvasir-SEG

Lab report · Alkhemyst-Ai

YassY Elhallaoui · 2026-08-25 · Pipeline: Auditing train/test leakage in a polyp benchmark: Kvasir-SEG (Jha et al. 2020, SimulaMet and the Cancer Registry of Norway)

Key findings

- 12 near-duplicate images were detected, forming 6 clusters.
- The largest near-duplicate cluster contained 2 images.
- The mean test leakage was 0.0056 with a standard deviation of 0.00697 across 25 random split repetitions.

Reproducibility

Everything needed to identify and repeat this run.

Field	Value
Run	run CT7bpz89SipR
Pipeline	Auditing train/test leakage in a polyp benchmark: Kvasir-SEG
Algorithm	Image Features
Started	2026-08-25 14:10:06
Finished	2026-08-25 14:12:07
Duration	120.9805 s
Compute	CPU
Figures	1

Data

The dataset used in this analysis is the Kvasir-SEG polyp benchmark, which consists of 1000 images. The dataset has 1 class and no duplicate rows were detected.

The analysis ran over 1,000 rows and 0 columns from kvasir-seg-images.zip.

Data quality

The dataset has 1 class and no duplicate rows were detected. However, the presence of near-duplicate images may indicate potential data quality issues.

Every step, in the order it ran

Each step below carries the settings it ran with, the numbers it recorded and the figures it produced, so a reader repeating this work can check one step at a time rather than the run as a whole.

1. kvasir-seg-images.zip

What it measured	Value
images	1000
classes	1

2. Image Features

backbone = resnet50, imageSize = 224, maxImages = 1000

What it measured	Value
feature dim	2048
backbone	resnet50
images	1000
pretrained	true

3. Near-Duplicate Audit

similarityThreshold = 0.9200, splitSeedCount = 25

What it measured	Value
near duplicate images	12
near duplicate rate	0.0120
near duplicate clusters	6
largest near duplicate cluster	2
similarity threshold	0.9200
random split repetitions	25
random split test leakage mean	0.0056
random split test leakage std	0.0070
random split straddling clusters mean	0.5600
recommendation	Candidate same-source image clusters cross the recorded rand

Results

The results showed 12 near-duplicate images, forming 6 clusters, with the largest cluster containing 2 images. The mean test leakage was 0.0056 with a standard deviation of 0.00697 across 25 random split repetitions.

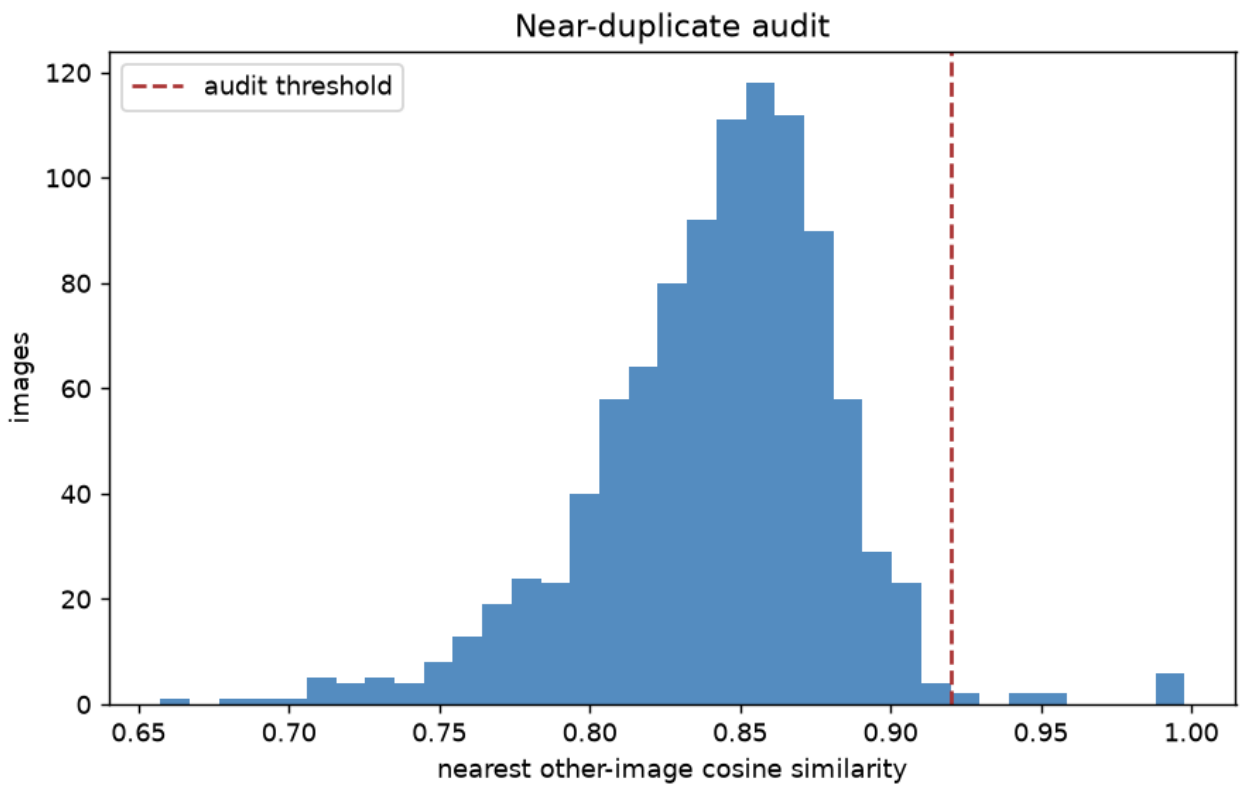

Figure 1: nearest similarity

Recorded metrics

The values below are copied directly from the run record; where the narrative cites a number, this table is authoritative.

Metric	Recorded value
n1787659832364 / images	1000
n1787659832364 / classes	1
n1787659846051 / feature dim	2048
n1787659846051 / backbone	resnet50
n1787659846051 / images	1000
n1787659846051 / pretrained	true
n kvasir near duplicate audit / near duplicate images	12
n kvasir near duplicate audit / near duplicate rate	0.0120
n kvasir near duplicate audit / near duplicate clusters	6
n kvasir near duplicate audit / largest near duplicate clust	2
n kvasir near duplicate audit / similarity threshold	0.9200
n kvasir near duplicate audit / random split repetitions	25
n kvasir near duplicate audit / random split test leakage me	0.0056
n kvasir near duplicate audit / random split test leakage st	0.0070
n kvasir near duplicate audit / random split straddling clus	0.5600
n kvasir near duplicate audit / recommendation	Candidate same-source image clusters cross the recorded rand

Limitations

The analysis is limited by the presence of near-duplicate images, which may indicate potential data quality issues. Additionally, the pipeline used a fixed similarity threshold of 0.92, which may not be optimal for all datasets.

Produced on the AlkhemYst-Ai platform. Every figure and every value in the tables of this report comes from the recorded run; the full experiment lineage is available in the workspace.